Anglo-Chinese Junior College

Physics Preliminary Examination
Higher 2

A Methodist Institution
(Founded 1886)

CANDIDATE NAME					CLASS			
CENTRE NUMBER	S	3	0	0	4	INDEX NUMBER		

PHYSICS **9478/02**

Paper 2 Structured Questions 28 August 2026
2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, class and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiners' use only	
1	/ 7
2	/ 7
3	/ 8
4	/ 11
5	/ 14
6	/ 8
7	/ 20
Total	/ 75

DATA AND FORMULAE

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{Gm}{r}$
temperature	T / K	=	T / °C + 273.15

pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}<c^2>$
mean translational kinetic energy of an ideal gas particle,	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t$
		=	$\pm\omega\sqrt{(x_0^2 - x^2)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \cdots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1}+\frac{1}{R_2}+\ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1}+\frac{1}{C_2}+\ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \cdots$
energy in a capacitor	U	=	$\frac{1}{2}QV=\frac{1}{2}\frac{Q^2}{C}=\frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1- e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n I$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

[Turn over

1 **(a)** State the conditions required for a system to be in equilibrium.

1 ……...

……..

2 ……...

……..…….
[2]

(b) Figure 1.1 shows a box hanging from a horizontal uniform beam of length 2.0 m. The beam is attached to the wall by a hinge and a cable at point P. Point M is the centre of mass of the beam.

The box has a mass of 4.0 kg, and the beam has a mass of 2.0 kg.

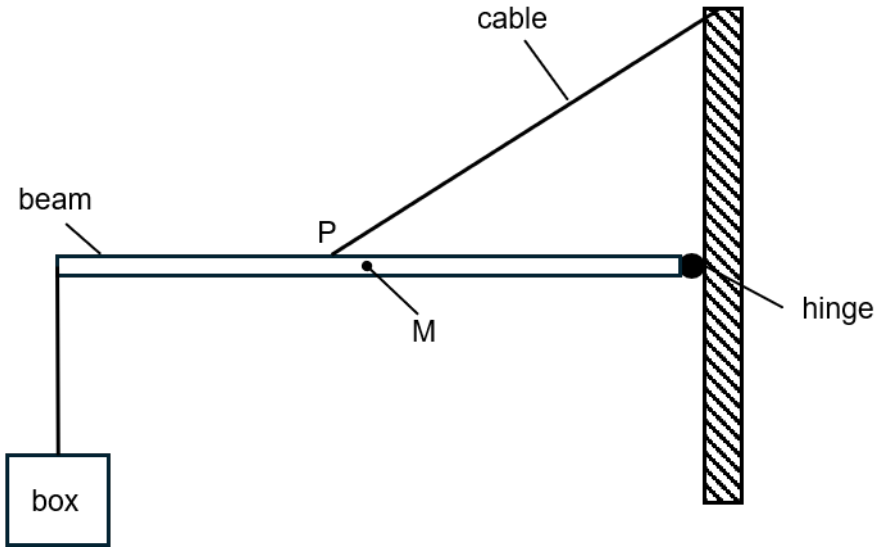


Figure 1.1

(i) On Figure 1.1, draw

1. the possible line of action of the combined weight of the beam and the box. Label the line L.
[1]

2. the force the hinge exerts on the beam. Label the force *H*. [1]

(ii) The box is now submerged in water, and the beam is maintained at

horizontal position.

State and explain the change, if any, to the angle that force *H* makes with the beam.

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[3]

[Total: 7]

2 A golf ball is hit from point A on the ground and moves through air to point B. The path

of the ball is shown in Figure 2.1.

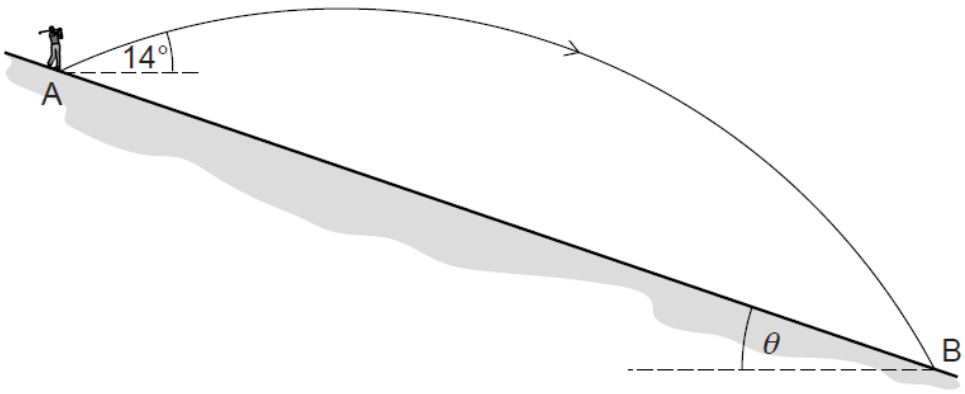


Figure 2.1

The ground slopes downhill and makes an angle θ to the horizontal as shown. The ball has an initial velocity of 63 m s^{-1} at an angle of 14° to the horizontal. It hits the ground at B after 4.9 s.

(a) Assuming air resistance to be negligible, calculate

(i) the vertical displacement from A to B,

vertical displacement = ... m [2]

(ii) the angle θ of the slope to the horizontal.

θ = ... ° [2]

(b) In a real situation, air resistance is not negligible.

(i) On Figure 2.1, sketch the path of the golf ball when air resistance is not negligible.
[2]

(ii) Explain how the maximum vertical height of the golf ball is affected.

……..

……..

……
[1]

[Total: 7]

For Examiner's Use

3 **(a)** A pellet is fired straight through a board that is 10.0 cm thick. It strikes the board with a speed of 120 m s^{-1} and emerges with a speed of 85 m s^{-1}.

Calculate

(i) the average deceleration of the pellet as it passes through the board,

deceleration = …… m s^{-2} [2]

(ii) the time taken for the pellet to pass through the board.

time taken = …… s [1]

(b) After passing through the board, the pellet in **3(a)** enters at a speed of 85 m s^{-1} into a wooden block that is initially at rest on a frictionless table.

The pellet remains in the block after impact.

(i) Explain why the principle of conservation of momentum can be applied in this collision.

……..

……..…….. [1]

(ii) Given that the masses of the pellet and the wooden block are 0.40 g and

50.0 g respectively, determine the speed of the block and pellet after impact.

speed = …… m s^{-1} [2]

(iii) Explain, with appropriate calculations, whether the collision is elastic or inelastic.

……..

…….. [2]

[Total: 8]

[Turn over

For Examiner's Use

4 A satellite of mass 1.0 kg orbits the Earth in a circular path.

Figure 4.1 shows the gravitational potential energy of the satellite at a distance r from the centre of the Earth in the region above the surface of the Earth. R_E is the radius of the Earth, which is 6400 km.

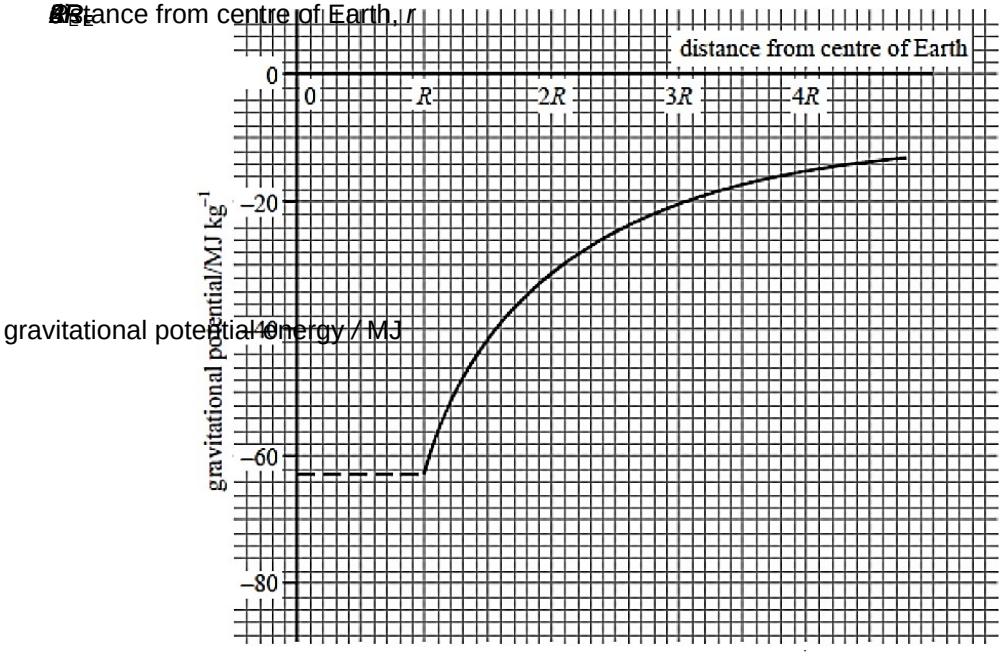


Figure 4.1

(a) (i) Define *gravitational potential* at a point.

……

……

……

……
[2]

(ii) Using Figure 4.1, deduce the gravitational potential at the Earth's surface.

gravitational potential = …… MJ kg^{-1} [1]

(b) **(i)** If a satellite of mass m is in an orbit of radius R around the Earth, show that its kinetic energy E_K is given by

$$E_K = \frac{GMm}{2R}$$

where M is the mass of the Earth and G is the gravitational constant.

[2]

(ii) On Figure 4.1, sketch the graph of the variation of the total energy E_T of the satellite of mass 1.0 kg with r for values of r greater than R_E.

Explain clearly how you obtained the graph.

……..

……..

……..

……..….
[3]

(c) **(i)** Write down an equation which relates gravitational force F_G, and gravitational potential energy U.

…….. [1]

For Examiner's Use

(ii) Using Figure 4.1, determine the gravitational force experienced by the satellite at a distance $2R_E$ from the centre of the Earth.

gravitational force = …… N [2]

[Total: 11]

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5 (a) Distinguish between electromotive force (e.m.f.) and potential difference (p.d.)

using energy considerations.

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……

…….. [2]

(b) A battery of e.m.f. 9.0 V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Figure 5.1.

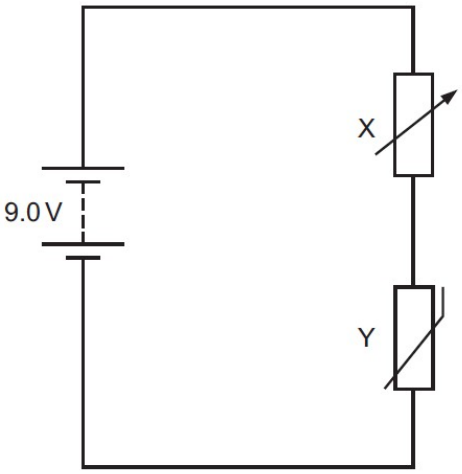


Figure 5.1

Figure 5.2 shows the variation with temperature of the resistance of the thermistor.

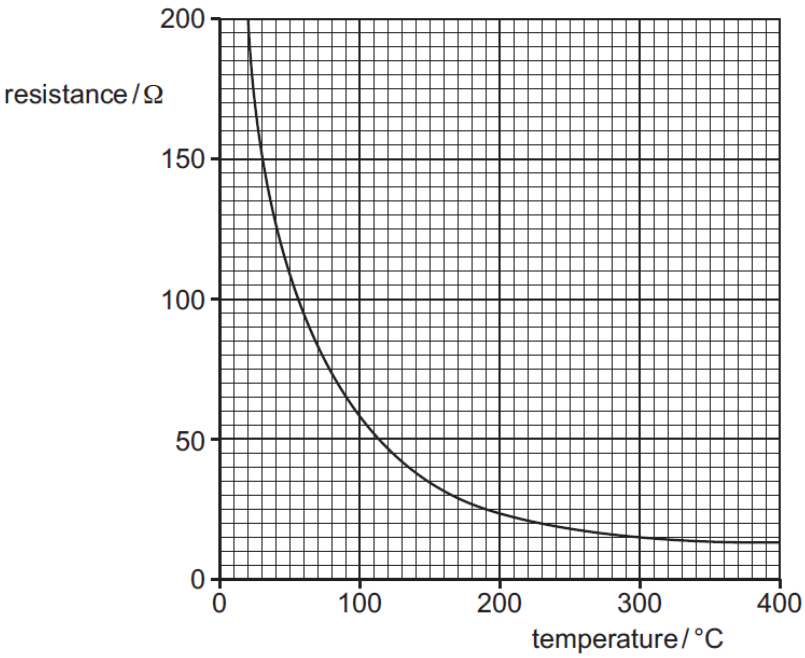


Figure 5.2

(i) The current in the circuit is 0.011 A and the potential difference across Y is

4.0 V.

Calculate the resistance of X.

resistance = …… Ω [2]

(ii) The temperature of Y is changed to 190 °C. The resistance of X remains unchanged.

1. Determine the new potential difference across Y.

potential difference = …… V [3]

2. The resistance of X is increased.

State and explain how the value in **5(b)(ii)1.** will change.

……..

……..

……..

……. [2]

(c) Figure 5.3 shows a network of capacitors connected to a 12 V supply. Z is a capacitor of capacitance *Z*.

[Turn over

For Examiner's Use

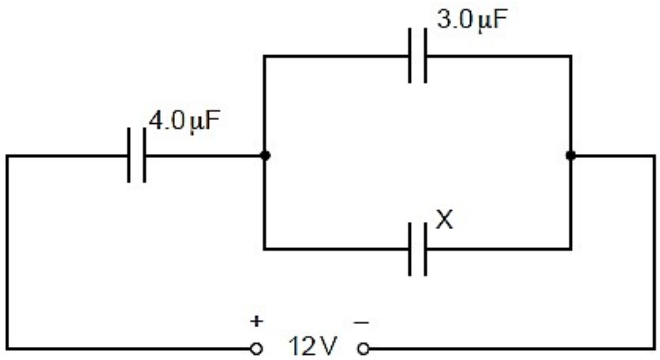


Figure 5.3

When the capacitors are fully charged, the potential difference across Z is 3.0 V.

(i) Calculate the charge stored in the 4.0 μF capacitor.

charge = …… C [2]

(ii) Determine the capacitance of Z.

capacitance = …… μF [3]

[Total: 14]

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6 An electron travelling horizontally with a velocity of 9.0×10^6 m s^{-1} enters perpendicularly into a uniform electric field between two parallel metal plates as shown in Figure 6.1. After leaving the electric field, it travels into a region of uniform magnetic field of magnetic flux density 0.72 mT. The magnetic field is directed normally into the plane of the paper.

Assume that viscous drag and gravitational force are negligible.

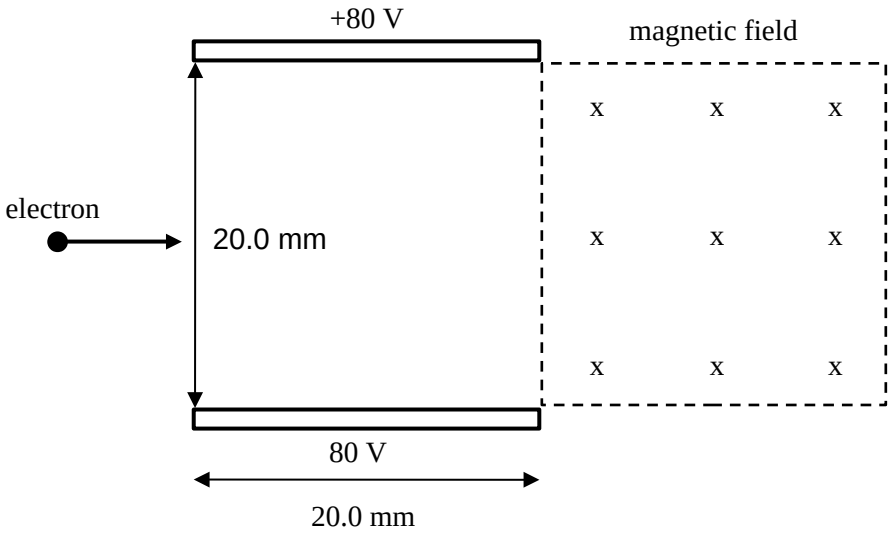


Figure 6.1

[Turn over

For Examiner's Use

(a) Determine the time taken for the electron to travel through the electric field.

time = …… s [1]

(b) Show that the velocity of the electron just before it enters the magnetic field is 9.5×10^6 m s^{-1}.

[3]

(c) Determine the radius of the path taken by the electron when it enters the magnetic field.

radius = …… m [2]

(d) On Figure 6.1, sketch a possible path taken by the electron in the electric and magnetic fields.
[2]

[Total: 8]

7 Read the passage and answer the questions that follow.

World Cup footballs have different panel shapes, seam patterns and surface textures. These features affect the airflow around the ball and hence the forces acting on it during flight.

Table 7.1 shows the physical properties of three official FIFA World Cup footballs released between 2010 and 2026.

Table 7.1

Ball (Year)	Number of Panels	Total Seam Length / m	Seam Width / mm	Seam Depth / mm
Jabulani (2010)	8	1.98	2.2	0.5
Al Rihla (2022)	20	3.52	5.8	1.6
Trionda (2026)	4	2.50	5.1	1.3

Figure 7.1 shows the seam and a panel of the Trionda.

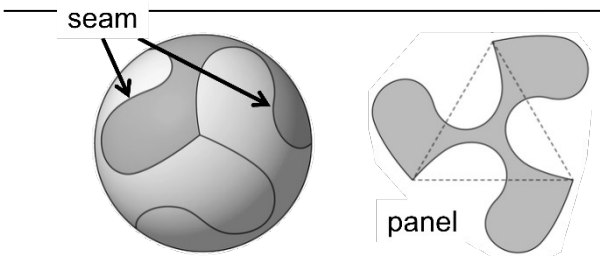


Figure 7.1

Modern FIFA World Cup footballs are engineered so that reducing the number of panels does not result in an excessively smooth surface, which could lead to undesirable changes in behaviour.

Although the Trionda has fewer panels than the Al Rihla, each panel contains three pronounced grooves that act as additional roughness elements. As a result, the Trionda has an overall effective surface roughness comparable to that of the Al Rihla. The two footballs therefore exhibit similar drag-crisis behaviour.

Wind-tunnel tests allow researchers to measure the drag coefficient C_D of a ball at different speeds.

The drag force F_D on a football is given by

$$F_D = \frac{1}{2} C_D \rho A v^2$$

where

[Turn over

For Examiner's Use

ρ : density of air
A : cross-sectional area of the ball
v : speed of the ball relative to the air

World Cup footballs are designed to have a mass of 0.430 kg and a diameter of 0.220 m to satisfy FIFA regulations. The density of air is taken to be 1.20 kg m^{-3}.

Figure 7.2 shows the variation with the speed v of the drag coefficient C_D for each football.

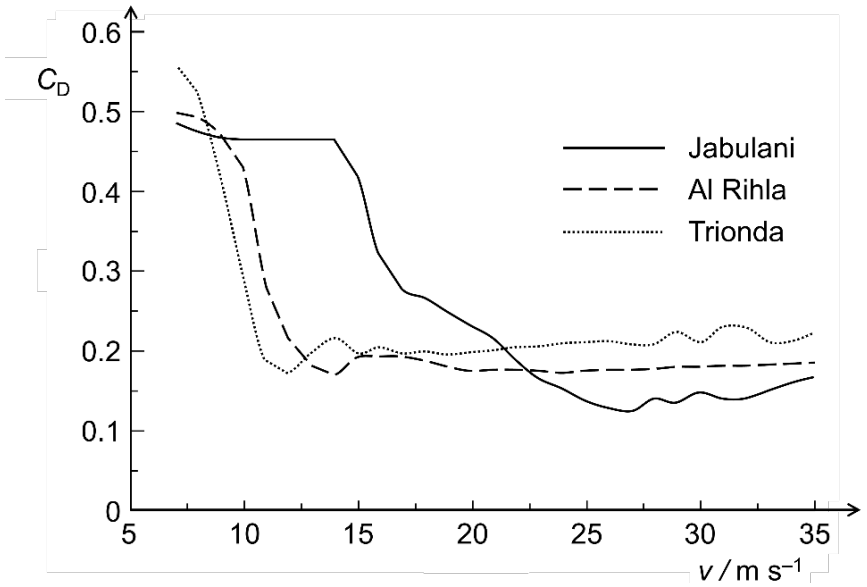


Figure 7.2

As the speed increases, the drag coefficient decreases rapidly over a range of speeds before becoming approximately constant. This behaviour is referred to as the drag-crisis region. The critical speed is defined as the speed at which the minimum drag coefficient occurs within the drag-crisis region. The critical speeds for the Trionda and Al Rihla are 11.9 m s^{-1} and 13.9 m s^{-1} respectively.

The flight of each football was simulated under identical launch conditions. Each football was kicked with the same initial speed, at the same launch angle of 25° and without spin. The predicted trajectories are shown in Figure 7.3.

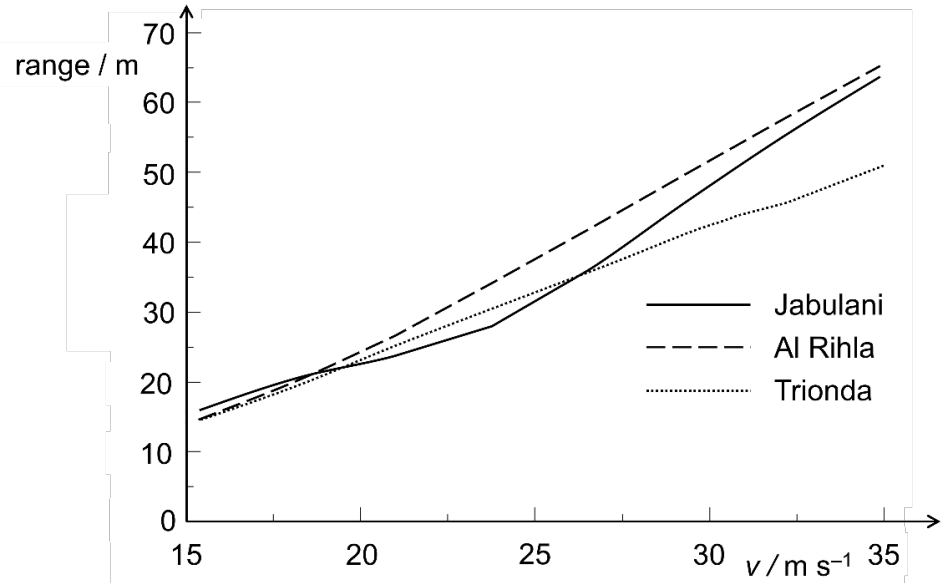


Figure 7.3

Match analysis indicates that short passes, used to maintain possession and accuracy between nearby teammates, are typically made at speeds between 12 and 15 m s^{-1}. Long kicks, used to move the ball over greater distances, are typically made at speeds exceeding 25 m s^{-1}.

(a) State the critical speed of the Jabulani.

critical speed = …… m s^{-1} [1]

(b) Using Table 7.1 and Figure 7.2, suggest why the Jabulani requires the highest speed to reach the drag-crisis region.

……..……..

……..……..

……..……..

……..……..

……..……..

……..……..
[3]

(c) Determine the drag force acting on the Jabulani at 30 m s^{-1}.

drag force = …… N [2]

For Examiner's Use

(d) The designers of the Trionda aimed to reduce drag during short passes while limiting the distance of long kicks.

Explain how the data show that these aims have been achieved.

……..……..

……..……..

……..……..

……..……..

……..……..

……..……..

[3]

(e) Both the Jabulani and Trionda travel at 30 m s^{-1}.

Determine the ratio

$$\frac{\text{rate of kinetic energy transferred to the surrounding air by the Jabulani}}{\text{rate of kinetic energy transferred to the surrounding air by the Trionda}}$$

ratio = …… [3]

(f) The coefficient of ballistic performance B indicates how important aerodynamic forces are on the trajectory of a ball in flight. It is given by

$$B = \frac{mg}{\rho A v^2}$$

where m is the mass of the ball.

(i) Derive an expression for the ratio $\frac{F_D}{mg}$ in terms of C_D and B.

[1]

(ii) Determine the value of $\frac{F_D}{mg}$ for the Trionda at a speed of 30 m s^{-1}.

$\frac{F_D}{mg}$ = ... [1]

(iii) Hence, state and explain whether air resistance is negligible for the Trionda at this speed.

.....

..... [1]

(iv) State and explain the trajectory of the ball when B tends to a large value.

.....

.....

.....

..... [2]

[Turn over

For Examiner's Use

(g) A spinning football also experiences an additional force known as the Magnus force. Figure 7.4 shows the airflow around a football spinning anticlockwise immediately after it leaves the player's foot and travels to the left.

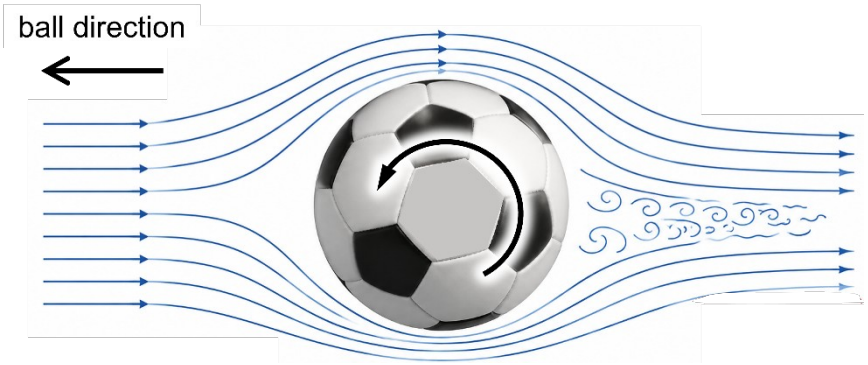


Figure 7.4

Experiments show that regions where the airflow speed is higher correspond to regions where the air pressure is lower.

(i) Explain why the top of the ball is at a higher pressure than the bottom of the ball.

……

……

…… [1]

(ii) The Magnus force causes the spinning football to curve with its speed remaining approximately constant over a short time interval.

State and explain the work done by the Magnus force on the ball.

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…… [2]

[Total: 20]

End of Paper